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Applications of AI in EEE

EEEE3134

Modern Programming

Python and VS Code

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What is a Programming Language?

Definition

- Used to communicate instructions to a computer. It allows developers to create software, automate tasks, process data, and solve real-world problems.

Why is it Important?

- Powers modern technology, from websites and mobile apps to AI and cloud services.
- Enables automation and improves efficiency across industries.
- Helps transform data into insights and supports decision-making.
- Serves as the foundation for software development and innovation.
- …

Most Widely Used Programming Languages

Language	Brief Description
Python	Easy to learn and widely used for AI, data science, automation, and web development.
JavaScript	The primary language of the web, used for interactive front-end and back-end applications.
Java	A reliable language commonly used for enterprise software and backend systems.
C++	A high-performance language used in games, systems, and real-time applications.
SQL	The standard language for storing, managing, and querying databases.
MATLAB	A numerical computing language widely used in engineering, research, and education.

Which programming language are you most familiar with?
How proficient are you in it, and have you completed any full projects using it?

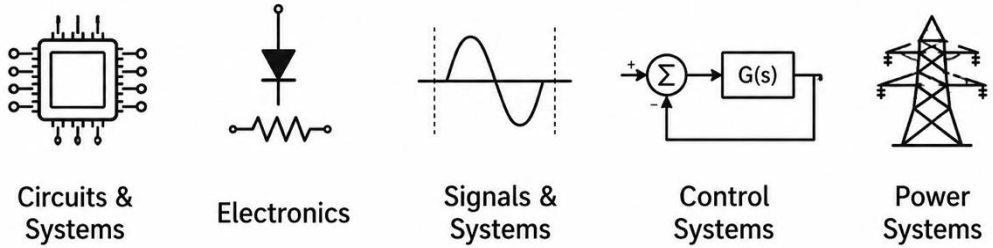
Programming Languages: Python

Why is it Popular	Key Features	Common Applications	Popular Libraries
• Easy to learn and use • Clear and readable syntax • Large open-source ecosystem • Strong community support • Widely adopted in industry and academia	• Interpreted and cross-platform • Supports multiple programming paradigms • Extensive standard library • Rich ecosystem of third-party packages • Rapid development and prototyping	• Artificial Intelligence • Machine Learning and Data Science • Web Development • Automation and Scripting • Scientific Computing • Cybersecurity	• NumPy Numerical computing • Pandas Data analysis • Matplotlib Data visualization • Scikit-learn Machine learning • TensorFlow/PyTorch Deep learning

Programming Languages: Matlab

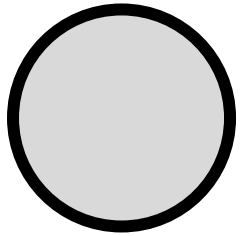
- **Core Strengths**
  - MATLAB is a programming language and computing platform for numerical analysis, data visualization, modeling, and simulation. Its built-in functions and specialized toolboxes enable efficient problem-solving in science and engineering.
- **Widely Used In**
  - MATLAB is widely used in engineering, research, and education, especially in *control systems, signal processing, robotics, communications, and aerospace*. Its integrated environment supports rapid prototyping, analysis, and visualization.

Electrical and Electronics Engineering



What Comes to Your Mind First

Draw a ball moving from the left to the right



Writing Codes: Traditional Way

```
while True:
    screen.fill(WHITE)          # erase old drawing

    ball_x = ball_x + speed     # move ball position

    pygame.draw.circle(
        screen,
        BLUE,
        (ball_x, ball_y),
        ball_radius
    )                           # draw ball in new position

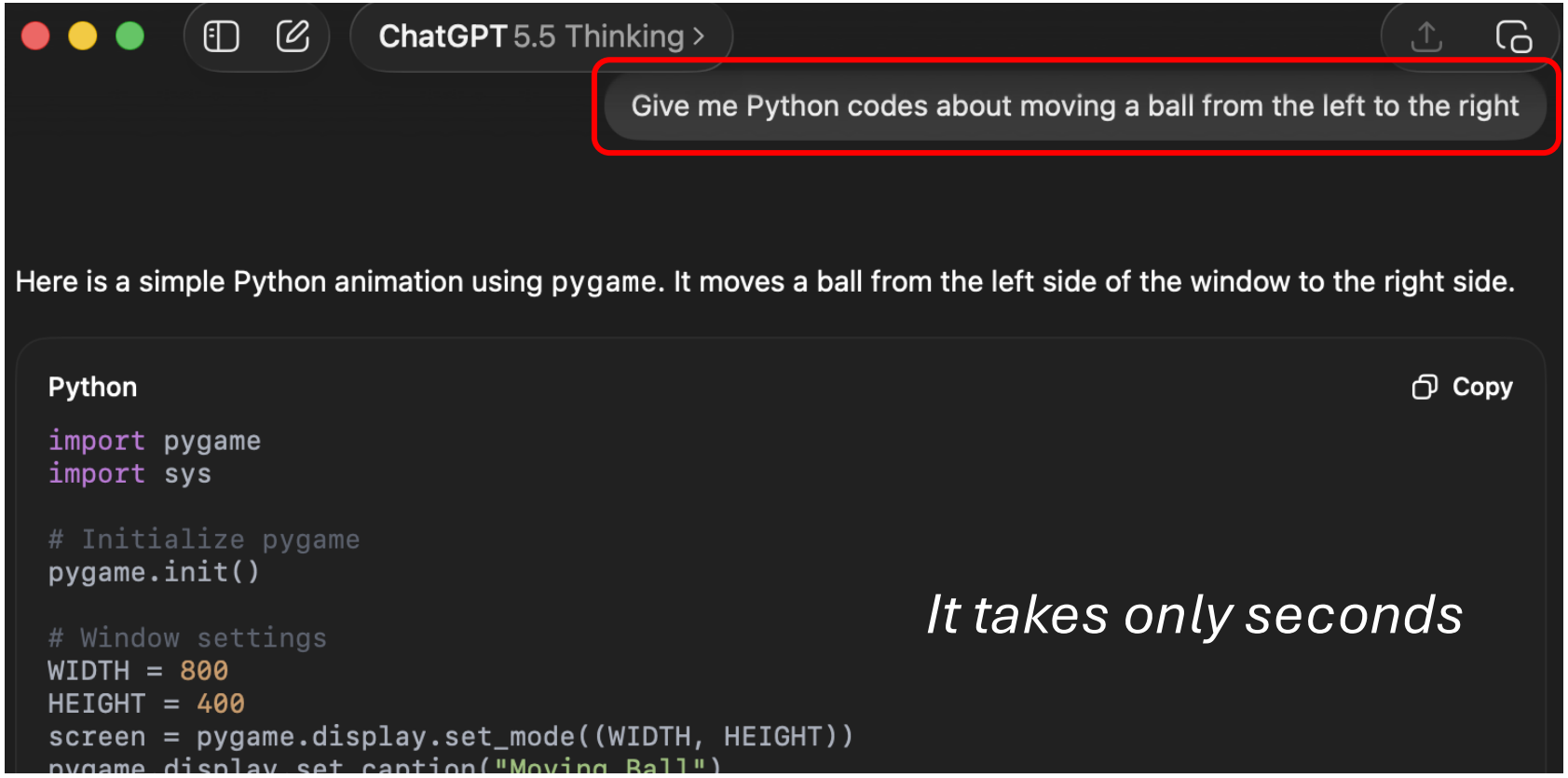
    pygame.display.flip()       # show the new frame
    clock.tick(60)              # control animation speed
```

Steps

1. Draw the ball at its current position.
2. Wait briefly.
3. Erase the old frame.
4. Change the ball's x-position.
5. Draw the ball again at the new position.
6. Repeat.

It may take hours years ago

Writing Codes: Today



It takes only seconds

ChatGPT

Claude

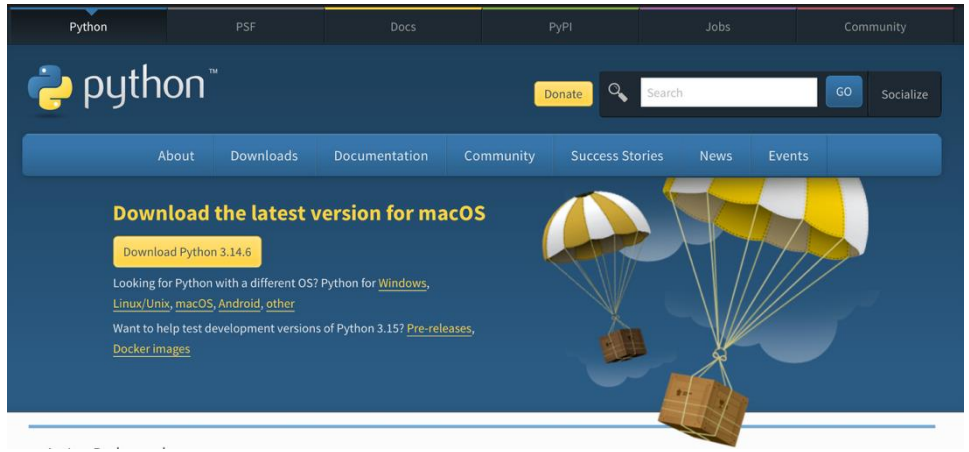
Gemini

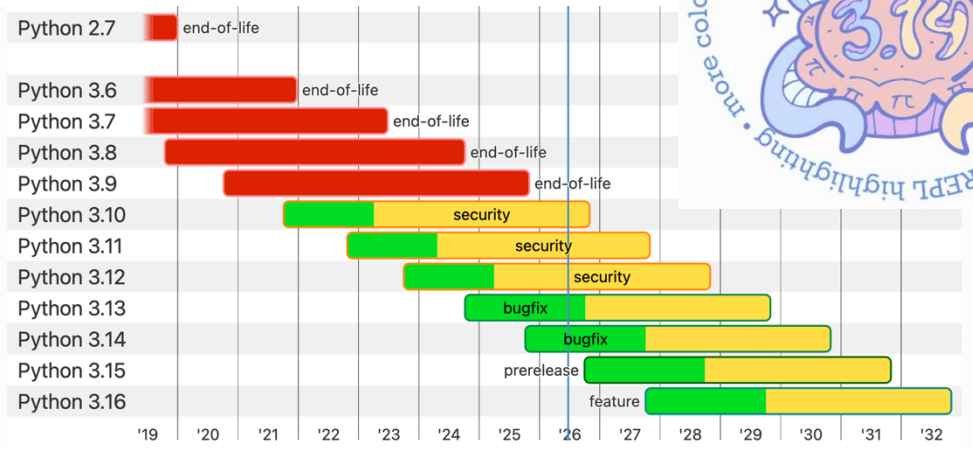
deepseek

For a New Computer w/ macOS

- **Install Python**
  - **Download and install** it from official website: *https://www.python.org/downloads*

    For beginners and new projects, choose the latest stable version (**v3.14**), not a pre-release version. If compatibility matters, use a widely supported stable version (v3.12 or v3.13).
  - **Verification** in **Terminal***: *python3 --version*





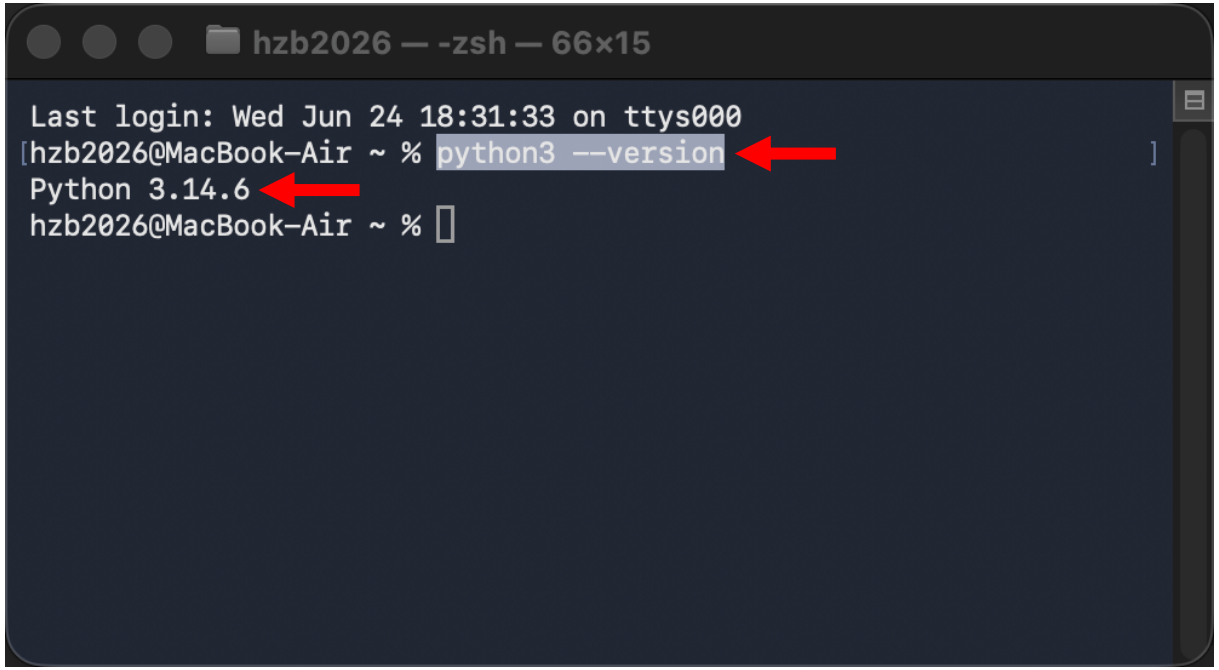


For a New Computer w/ macOS

- **Terminal**
  - A **text-based interface** that allows users to interact directly with the operating system by entering commands.
  - **Importance**
    - Historically, the terminal was important because it provided efficient access to operating systems and development tools.
    - *In the AI era, **its importance has grown further** because AI assistants (say ChatGPT) interact most naturally through **text-based** commands and scripts.*



Terminal

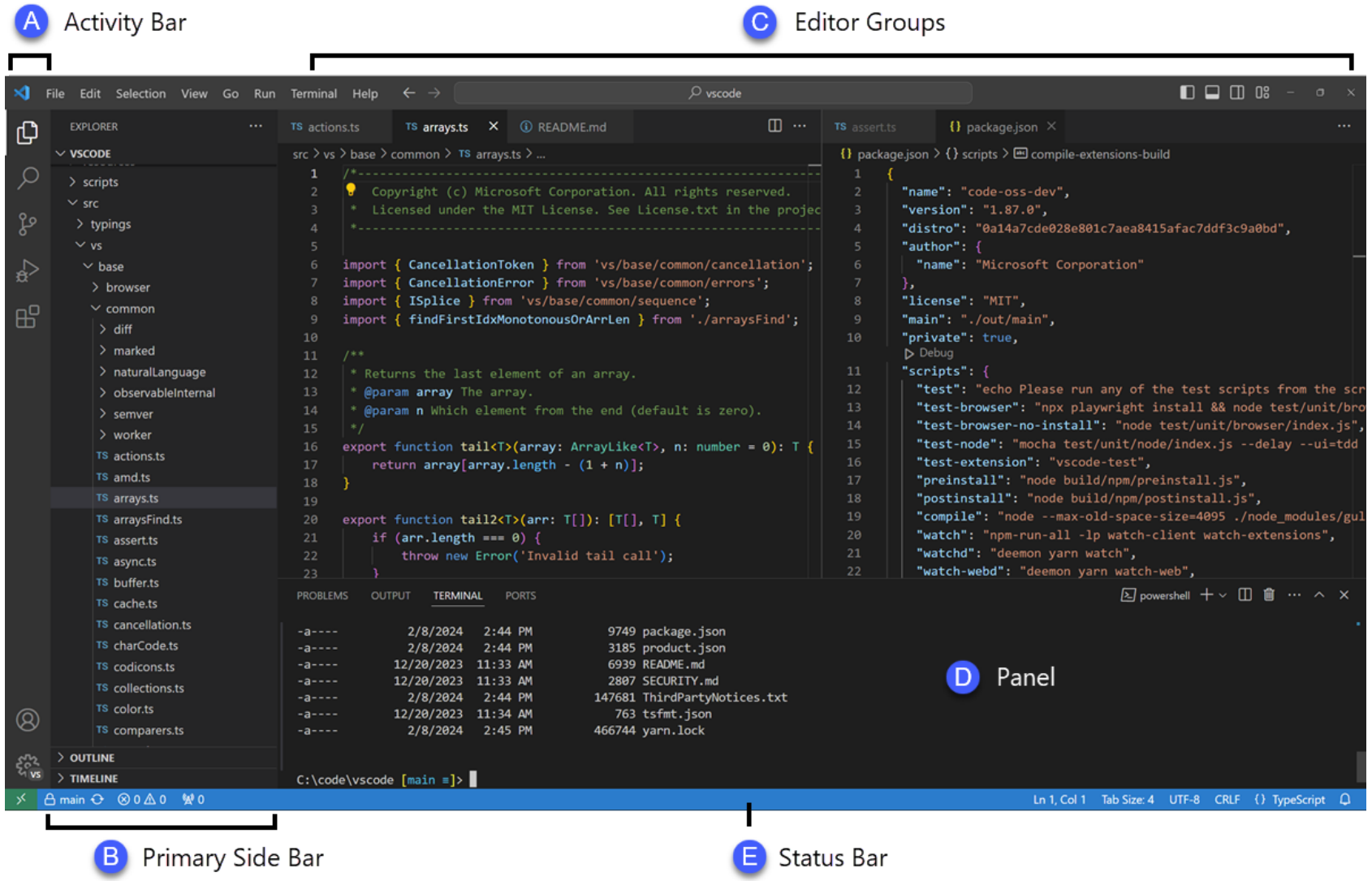


Integrated Development Environment (IDE)

Language	Brief Description
Visual Studio Code	The most popular lightweight code editor today. Supports almost every programming language through extensions.
Visual Studio	A full-featured IDE primarily used for C#, .NET, C++, and Windows application development.
PyCharm	A professional IDE designed specifically for Python development, with excellent debugging and data science support.
Eclipse	A long-established open-source IDE, widely used in Java and enterprise software development.
Jupyter Notebook	Widely used for data science, machine learning, and education, allowing code, text, and visualizations in one document.
MATLAB	A numerical computing language widely used in engineering, research, and education.

Which IDEs are you most familiar with?

IDE: Visual Studio Code (VS Code)



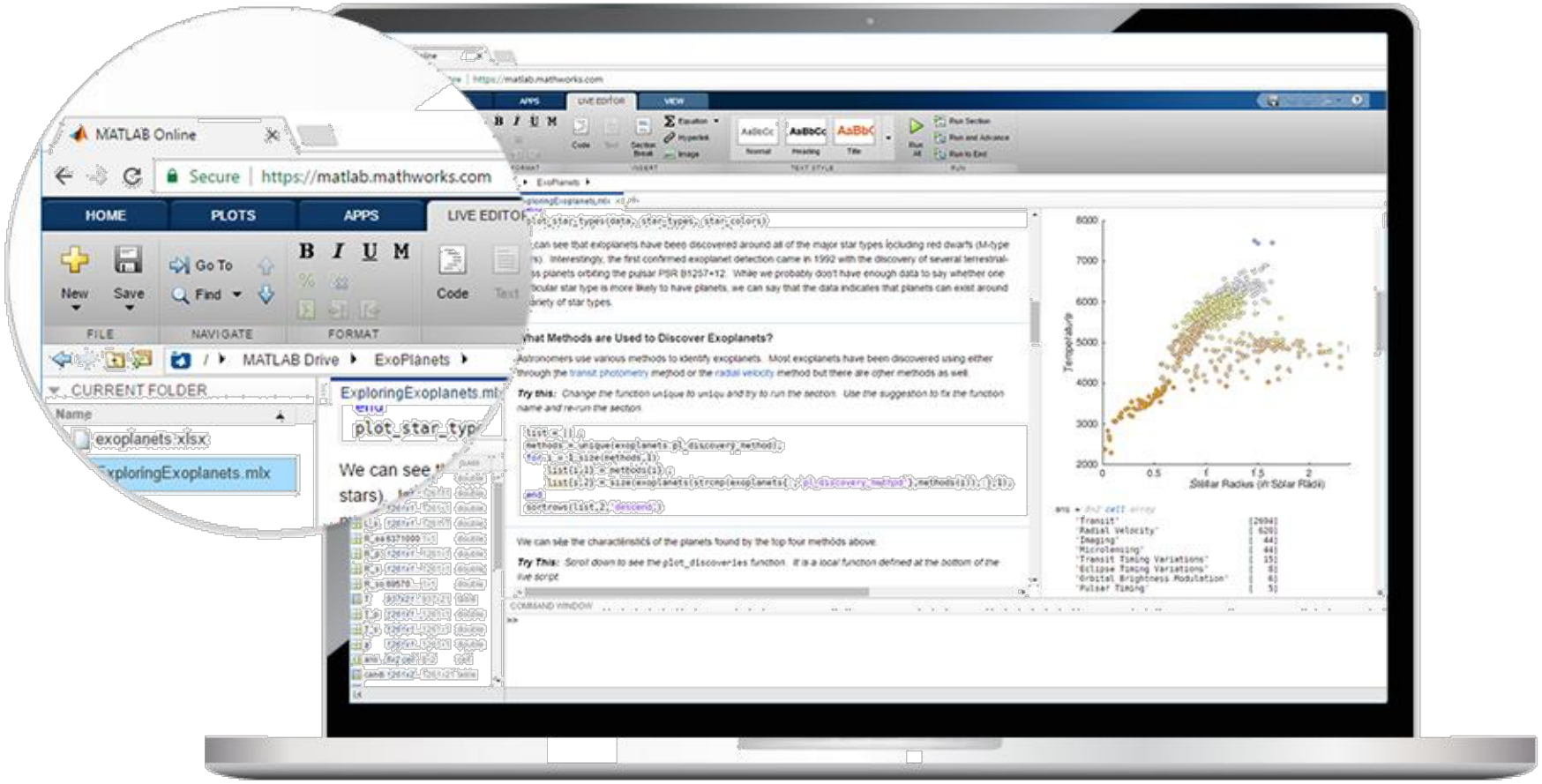
Visual Studio Code

IDE: Visual Studio Code (VS Code)



Why is it Popular	Key Features	Commonly Used For
• Free and open-source • Available on Windows, macOS, and Linux • Supports dozens of programming languages • Large extension marketplace • Fast, lightweight, and easy to customize	• Syntax highlighting and Intelligent sense • Built-in debugging tools • Integrated terminal • Git and GitHub integration • AI-assisted coding support (e.g., ChatGPT, GitHub Copilot)	• Python development • Web development (HTML, CSS, JavaScript, TypeScript) • C/C++ programming • Data science and machine learning • General software development

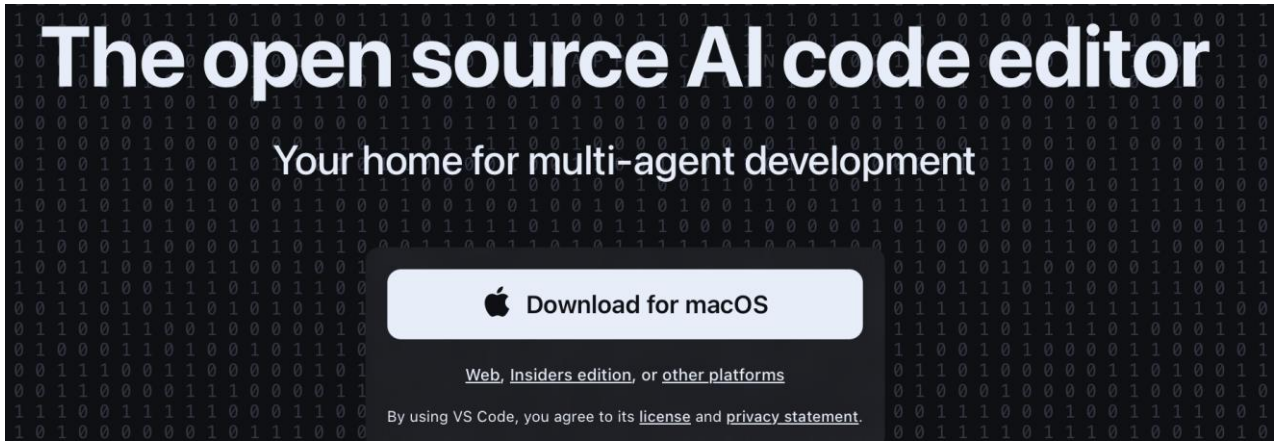
IDE: Matlab



MATLAB

For a New Computer w/ macOS

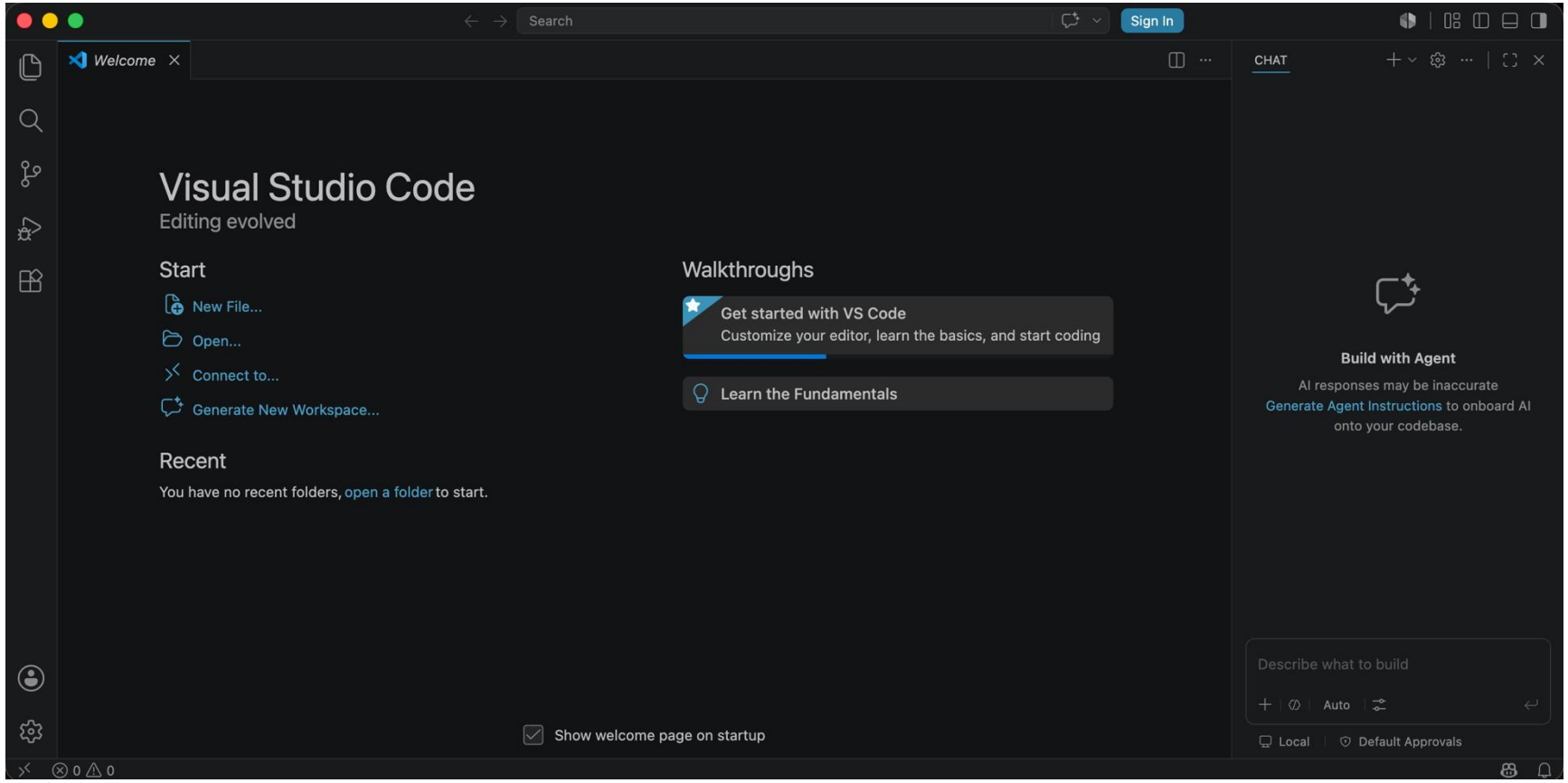
- **Install VS Code**
  - **Download** and **install** it from official website: *https://code.visualstudio.com*





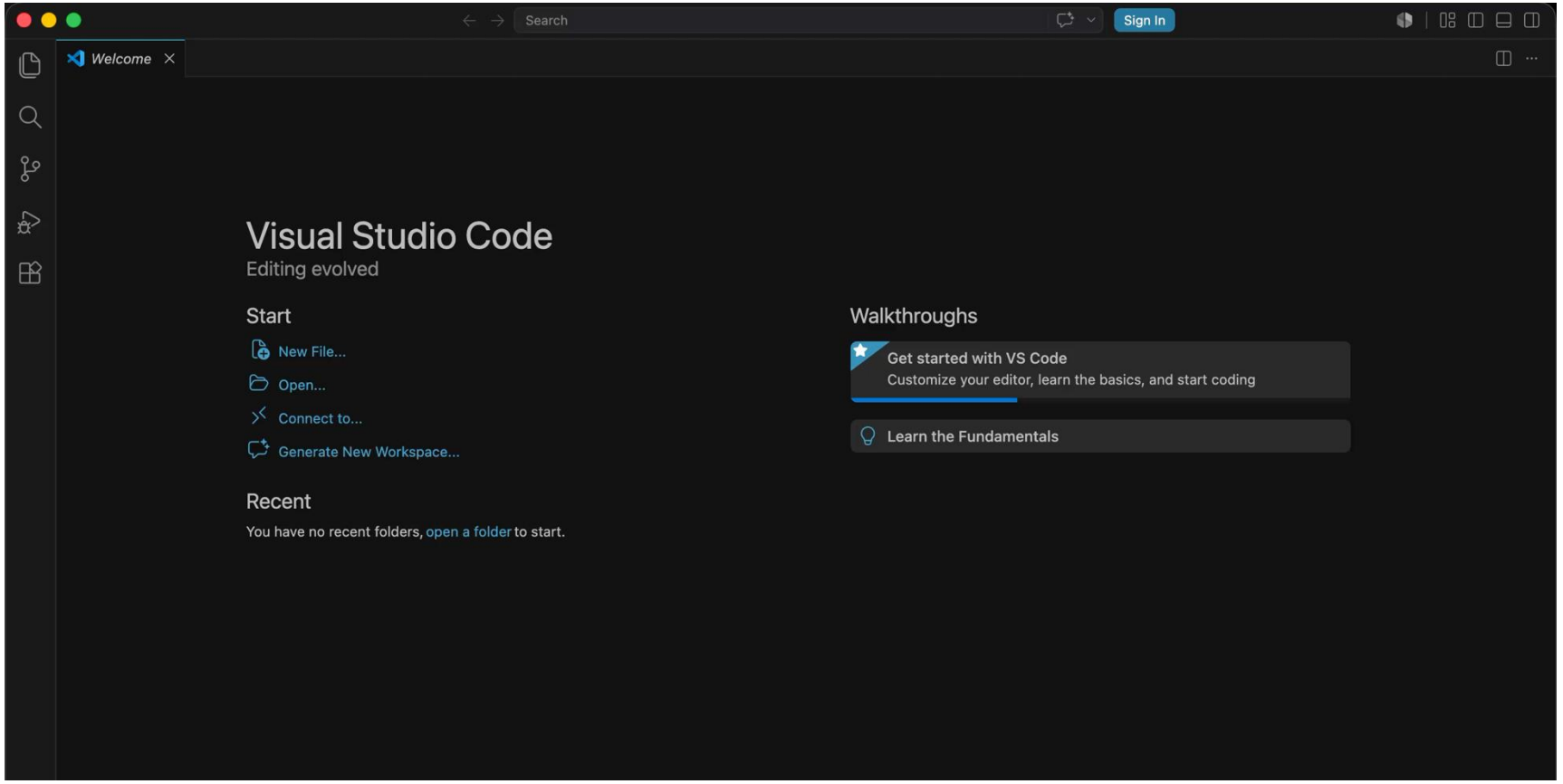
For a New Computer w/ macOS

- **Install VS Code**
  - **Download** and **install** it from official website: *https://code.visualstudio.com*



For a New Computer w/ macOS

- **Install VS Code**
  - **Test** by writing the first Python file



Thank You

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